Response-Based Derivation of MAAT Structural Weights

From Covariance Geometry to Selection Pressure

Christof Krieg

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Abstract

The MAAT structural-selection framework uses sector weights λ_a to quantify the relative contribution of primitive defect sectors $(d_H, d_B, d_S, d_V, d_R)$ to an effective structural functional. Previous benchmarks calibrated these weights by maximum-entropy fitting. This paper provides the next closure step: it replaces purely heuristic weight fitting by a covariance-based response derivation, establishing the first effective physical closure of the MAAT selection-weight sector.

Starting from an empirical reference measure μ_0 and an exponential reweighting of the ensemble,

$$P_\lambda[X] \propto \mu_0[X] \exp[-\lambda \cdot d(X)],$$

linear response gives

$$\delta\langle d_a\rangle = -\sum_b \mathrm{Cov}_{\mu_0}(d_a, d_b)\lambda_b.$$

Thus the structural weights are obtained as response pressures:

$$\lambda = \left(C + \eta\,\frac{\mathrm{tr}(C)}{A}\mathbb{I}\right)^{-1} \left(\langle d\rangle_{\mu_0} - \langle d\rangle_*\right).$$

Applied to the fused SAT/MAAT-Core/boundary ensemble, the closure naturally recovers robustness dominance when safety and boundary information are included in the target. This provides an effective statistical derivation of the weights as response pressures, without assuming universal microscopic constants.

1 Motivation

In previous MAAT and structural-selection papers, the sector weights λ_a entered the effective functional

$$\mathcal{F}_{\mathrm{struct}}[X] = -\sum_a \lambda_a \log(\epsilon + \Gamma_a[X]), \qquad \Gamma_a[X] = \frac{1}{1 + d_a[X]}. \tag{1}$$

The weights were interpreted as selection pressures, but their operational origin remained partly open. Maximum-entropy fitting already identifies them as Lagrange multipliers, analogous to inverse temperature in statistical mechanics [1, 2]. This was the role of the previous MAAT weight-calibration and constants benchmarks [9, 10]. The response-based closure is the next logical step because it asks not only which multipliers reproduce a target distribution, but which multipliers are generated by the fluctuation geometry of the reference ensemble itself. The remaining question is therefore sharper:

Can the weights be computed from fluctuations of the defect ensemble itself?

The contribution of this paper is to answer this question conditionally: given a reference ensemble μ_0, primitive defect definitions, and a target selection condition, the sector weights can be computed as response pressures rather than freely chosen fit parameters.

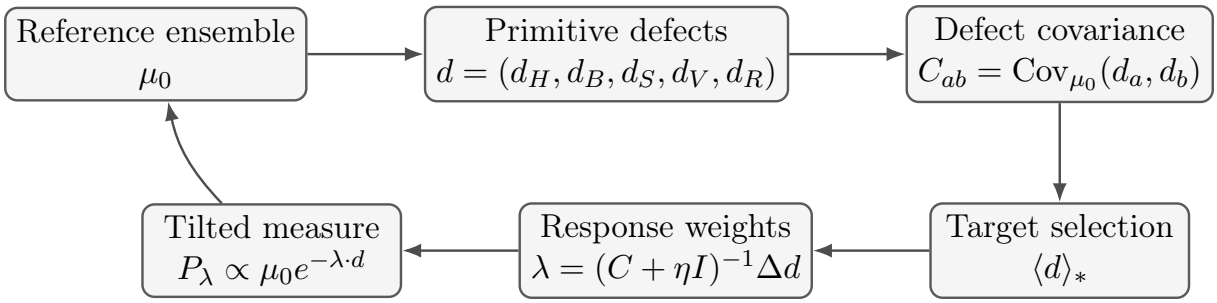

Figure 1: Closure workflow. The sector weights are obtained from the covariance geometry of primitive defects and a specified target selection condition. The result is an effective response pressure, not a universal constant.

2 Reference Measure and Exponential Tilting

Let X denote a configuration, state, instance, or candidate structure, and let

$$d(X) = (d_H, d_B, d_S, d_V, d_R) \tag{2}$$

be its primitive MAAT defect vector. The empirical reference measure μ_0 represents the unselected ensemble. Selection is introduced by exponential tilting, equivalently an exponential reweighting of the reference measure by structural defects. This is the standard form associated with constrained entropy maximisation and information projection [3, 4]:

$$P_\lambda[X] = \frac{1}{Z(\lambda)} \mu_0[X] \exp[-\lambda \cdot d(X)]. \tag{3}$$

The partition function is

$$Z(\lambda) = \int \mathrm{d}\mu_0(X) \, \exp[-\lambda \cdot d(X)]. \tag{4}$$

The expected defect vector under the tilted measure is

$$\langle d_a \rangle_\lambda = -\frac{\partial \log Z}{\partial \lambda_a}. \tag{5}$$

At the level of ensembles, the same structure can be read as an effective rate-function construction in the language of large deviations [7].

3 Linear Response Closure

The closure used here is a *linear-response approximation*. It is valid when the selected ensemble can be represented as a moderate exponential tilt of the reference ensemble. Strongly nonlinear target shifts may require higher-order cumulants or a full non-linear maximum-entropy solve. The use of covariance as the first response kernel follows the same structural logic as classical fluctuation-response theory [5, 6].

Expanding around $\lambda = 0$ gives

$$\left. \frac{\partial \langle d_a \rangle_\lambda}{\partial \lambda_b} \right|_{\lambda=0} = -\mathrm{Cov}_{\mu_0}(d_a, d_b). \tag{6}$$

Therefore

$$\langle d \rangle_\lambda \approx \langle d \rangle_{\mu_0} - C\lambda, \qquad C_{ab} = \mathrm{Cov}_{\mu_0}(d_a, d_b). \tag{7}$$

Given a selected target ensemble with mean defect vector $\langle d \rangle_*$, the response equation becomes

$$C\lambda \approx \langle d \rangle_{\mu_0} - \langle d \rangle_*. \tag{8}$$

With ridge regularisation,

$$\boxed{\lambda = \left(C + \eta \frac{\mathrm{tr}(C)}{A}\mathbb{I}\right)^{-1} (\langle d\rangle_{\mu_0} - \langle d\rangle_*)} \tag{9}$$

where $A = 5$ is the number of sectors. The ridge term stabilises the inversion of the covariance matrix. This is important because primitive defects are not statistically independent; strong correlations can make C poorly conditioned even when the underlying ensemble is well-defined. The regularisation plays the same numerical role as Tikhonov regularisation for ill-conditioned inverse problems [8]. Since λ_a is interpreted as a non-negative penalty weight, negative raw components are projected to zero. A negative raw component means that the corresponding sector is not acting as a penalty for the chosen target; it is therefore treated as inactive in the positive penalty representation rather than interpreted as an anti-constraint.

4 Dataset and Target Ensembles

The closure is evaluated on the fused boundary-aware defect ensemble used in the previous MAAT weight-calibration benchmark:

Source	Samples	Role
SAT hardness atlas	2000	combinatorial constraint instances
MAAT-Core	500	unconstrained optimisation states
MAAT-Core boundary	900	safe, near-boundary, and violated regimes
Total	3400	fused defect ensemble

Four target definitions are used:

Target	Meaning
low-defect 20%	generic low-defect structural selection
safe + core-safe	explicitly safe and unconstrained-safe states
safe boundary only	safe states from the boundary ensemble
not violated	all configurations except explicit violations

5 Results

Using $\eta = 10^{-3}$, the response-derived normalised weights are:

Target	H	B	S	V	R
low-defect 20%	0.1587	0.2099	0.1808	0.1534	0.2972
safe + core-safe	0.0945	0.0701	0.0000	0.0023	0.8331
safe boundary only	0.1093	0.1830	0.1911	0.1597	0.3569
not violated	0.0000	0.0000	0.2602	0.0000	0.7398

The main qualitative result is:

> *Robustness dominance emerges as a covariance-response effect of safety/boundary target selection.*

Thus the dominance of R is not merely imposed by hand. It follows from how the target selection condition deforms the reference defect distribution.

The near-zero weights in some target definitions also carry information. For the "safe + core-safe" target, the dominant task is to suppress robustness defects, so S and V become effectively

inactive in the positive penalty representation. For the "not violated" target, excluding explicit failures mainly activates R and, secondarily, S; the other sectors do not contribute as independent penalty directions after covariance correlations are accounted for.

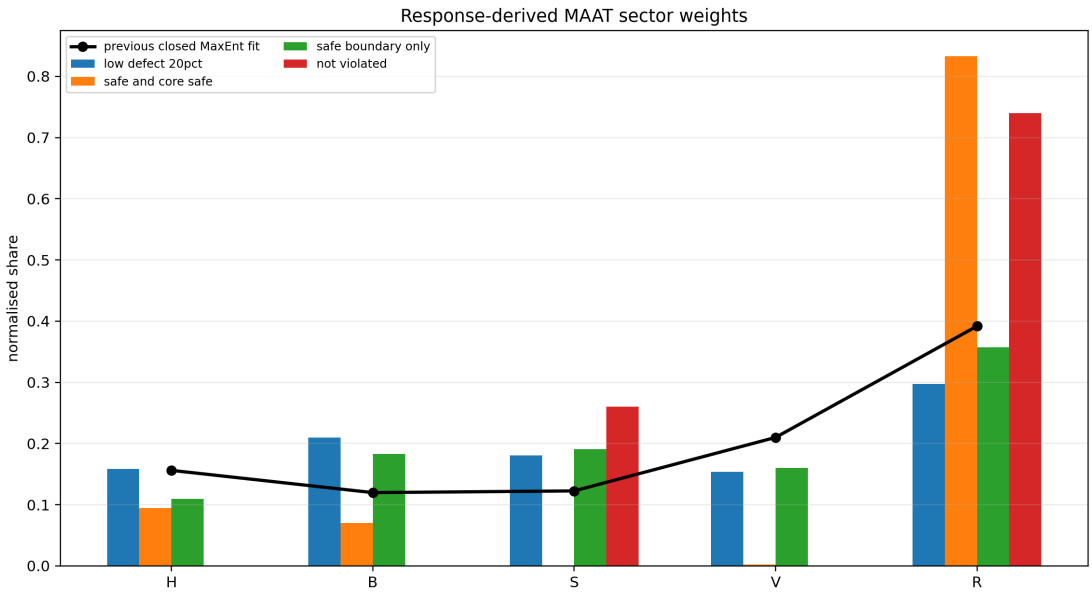

Figure 2: Response-derived sector shares for different target ensembles. The previous closed MaxEnt fit is shown as a black reference curve.

6 Interpretation

Equation (9) gives a precise meaning to the sector weights:

$$\lambda_a \sim \text{response pressure conjugate to defect } d_a. \tag{10}$$

This is directly analogous to thermodynamic response coefficients. In ordinary maximum entropy, inverse temperature is the Lagrange multiplier conjugate to energy. Here the λ_a are Lagrange multipliers conjugate to primitive structural defects.

The closure also clarifies why there is no single universal value of λ_a at the present effective-theory level. The weights depend on:

- the reference measure μ_0,
- the chosen primitive defect definitions,
- the covariance structure of the ensemble,
- the target selection condition.

This dependence is not a weakness; it is the expected behaviour of effective response coefficients.

7 What Has and Has Not Been Derived

Derived:

- λ_a can be obtained from defect covariance and target response.
- the weights are no longer arbitrary free parameters in a fixed ensemble.

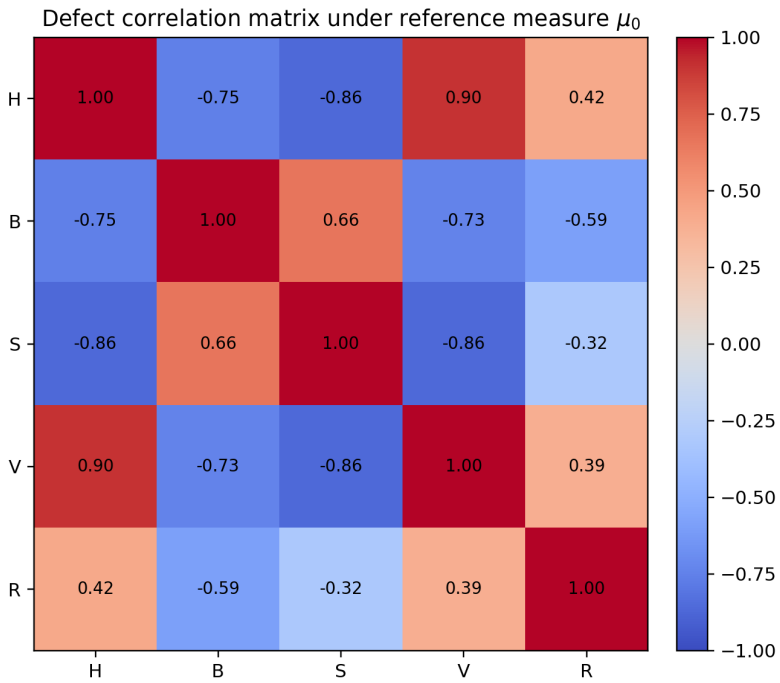

Figure 3: Correlation matrix of primitive defects under the empirical reference measure μ_0. The response weights are determined by this covariance geometry together with the selected target defect vector.

- robustness dominance follows from boundary/safety selection geometry.

Not derived:

- universal microscopic constants of nature,
- a unique measure over all possible physical configurations,
- a first-principles derivation from a fundamental action such as string theory or quantum gravity.

8 Reproducibility

The full reproducibility bundle is stored in:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/lambda_response_closure

Run:

```
cd experiments/lambda_response_closure
python3 lambda_response_closure.py
```

This generates the result JSON and all figures used in this note.

9 Conclusion

The response closure provides the first effective physical derivation of MAAT sector weights from ensemble fluctuations. It does not turn λ_a into universal constants, but it does remove their purely heuristic status: within a specified ensemble and target condition, they are computable response pressures.

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